

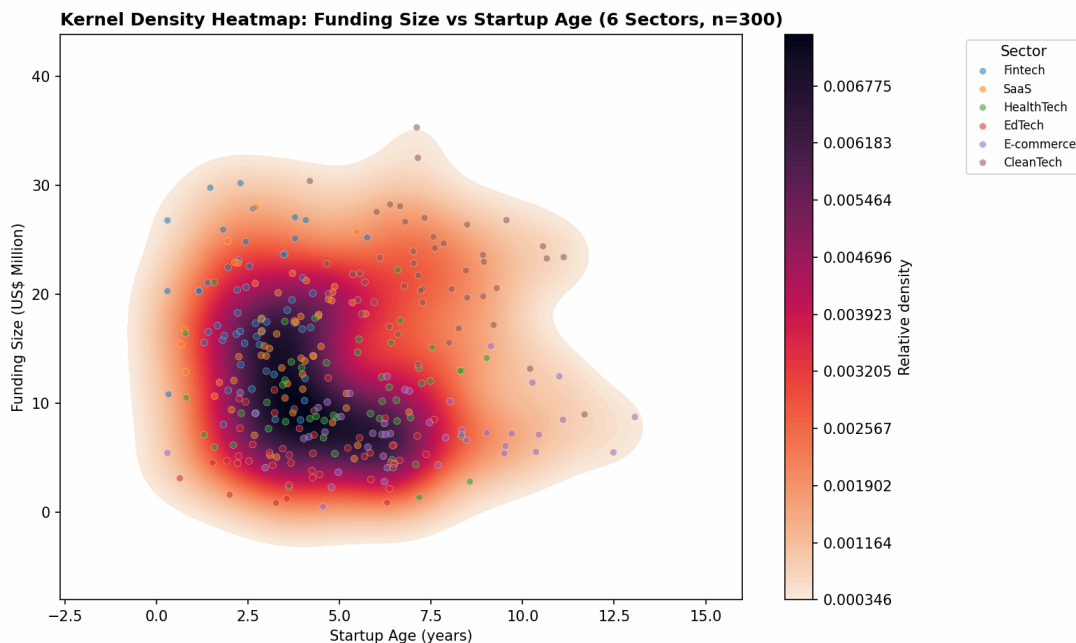
Data Visualization & Analysis Assignment

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Q1. Geospatial Visualization — KDE Heatmap (Funding Size vs Startup Age)

Brief: A VC firm's Investment Chair wants to study funding size vs startup age across 6 sectors (Fintech, SaaS, HealthTech, EdTech, E-commerce, CleanTech). A 300-startup dataset was simulated with sector-specific age/funding distributions, and a 2D kernel-density heatmap was built over (age, funding).

```
for sec,(am,astd,fm,fstd,n) in sectors.items():
    age = clip(normal(am, astd, n), 0.3, 15)
    fund = clip(normal(fm, fstd, n), 0.5, None)
    rows.append(DataFrame({'sector':sec,'age':age,'funding':fund}))
df = concat(rows)
sns.kdeplot(df, x='age', y='funding', fill=True, cmap='rocket_r', levels=100)
sns.scatterplot(df, x='age', y='funding', hue='sector')
```



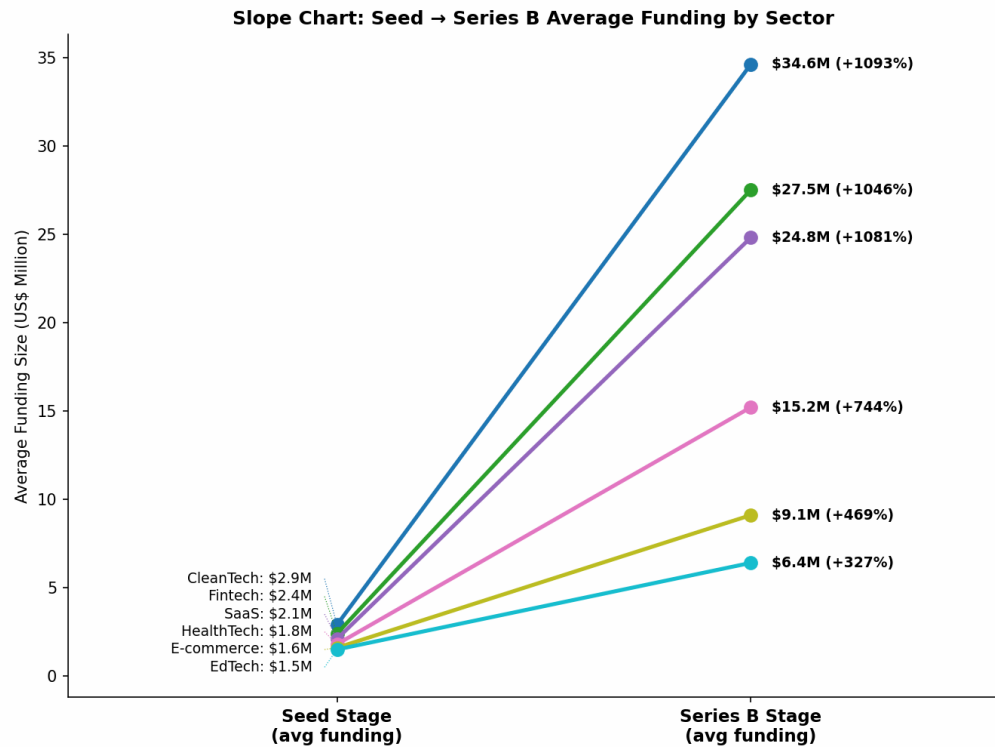
Analysis

- **Single dominant hotspot:** the darkest (near-black) core of the density surface sits at age $\approx 3.5\text{--}4$ yrs and funding $\approx \$9\text{--}15\text{M}$, with the single highest-density grid point at age=3.83 yrs, funding=\$9.43M (peak KDE density ≈ 0.0074 on the 0–0.0068+ scale).
- **Sector overlap explains the hotspot:** the sample means of three sectors fall directly inside this hot core — Fintech (2.65 yrs, \$18.50M), SaaS (3.58 yrs, \$15.19M) and HealthTech (5.14 yrs, \$10.77M) — so the hotspot is really a 3-sector overlap zone, not one sector alone.
- **Secondary, weaker cluster:** CleanTech (mean 7.34 yrs, \$21.27M) forms a lighter, more spread-out (orange, not black) region in the upper-right of the map, extending out to ages 10–13 yrs — density here is visibly lower than the core, i.e. more heterogeneous / less consistent deal-making.
- **Cold zone:** EdTech (4.33 yrs, \$5.46M) and E-commerce (6.93 yrs, \$7.52M) sit in the pale lower band of the map (funding well under \$10M), outside the hot core entirely.
- **Known driver alignment:** the 2–5 yr / \$8–20M hotspot matches the standard Seed→Series A capital-raise cadence in venture funding, explaining why software/services sectors (Fintech, SaaS, HealthTech) cluster there, while CleanTech's older/larger-ticket cluster reflects its capital-intensive, longer product-development cycle.

Q2. Relationships & Multivariate — Paired/Slope Chart (Seed vs Series B)

Brief: Using the same 6-sector data, a paired before-vs-after slope chart compares each sector's average funding at Seed stage vs Series B stage.

```
df['change'] = df.seriesB_musd - df.seed_musd
df['pct_change'] = df.change/df.seed_musd*100
for i,row in df.iterrows():
    ax.plot([0,1],[row.seed_musd,row.seriesB_musd], marker='o')
    ax.text(1.05,row.seriesB_musd,f'$ {row.seriesB_musd}M (+{row.pct_change:.0f}%)')
```



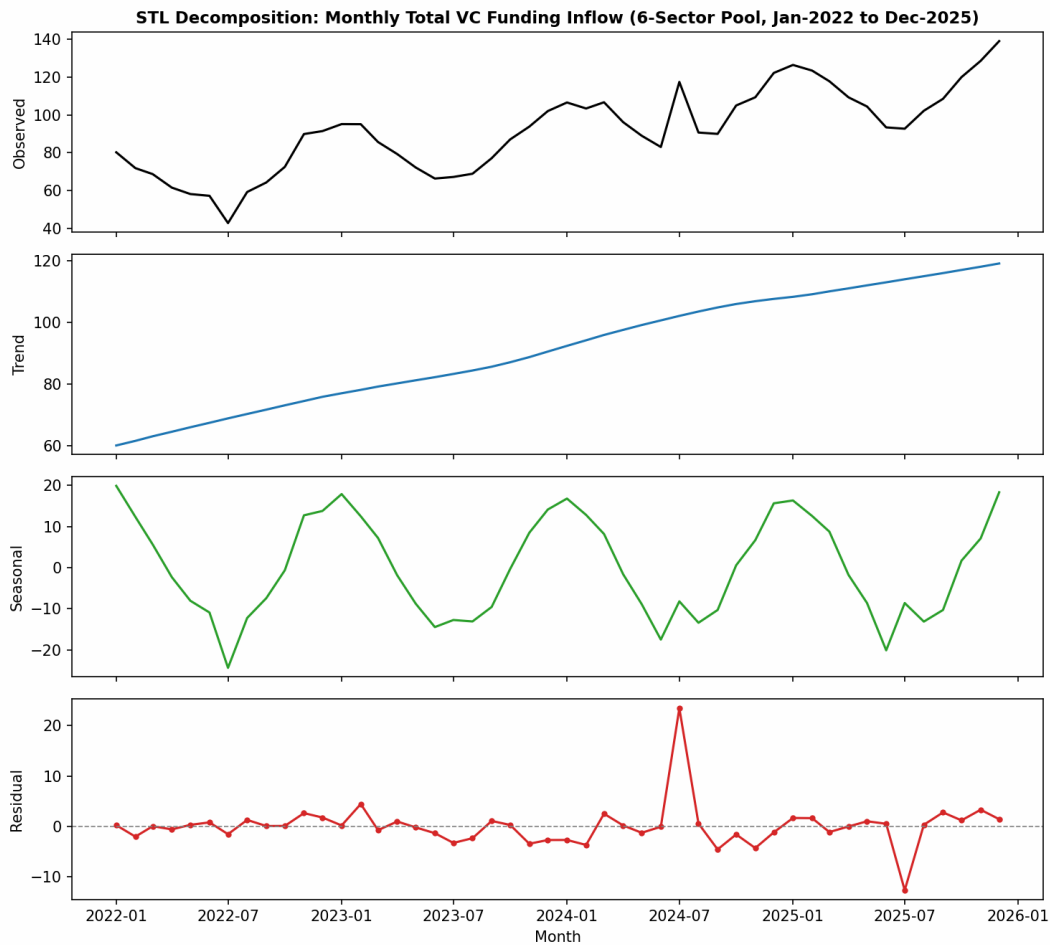
Analysis

- **Largest absolute improvement:** CleanTech, \$2.9M → \$34.6M (+\$31.7M, +1093%) — the steepest line on the chart and the largest jump of all six sectors.
- **Close second/third:** SaaS grew \$2.1M → \$24.8M (+1081%) and Fintech \$2.4M → \$27.5M (+1046%) — all three “hot” sectors cross the \$1000% growth mark.
- **Moderate improvement:** HealthTech rose \$1.8M → \$15.2M (+744%), clearly ahead of the bottom two but well behind the top three.
- **Sectors that “stayed flat” in relative terms:** E-commerce (+469%, ending at just \$9.1M) and EdTech (+327%, ending at only \$6.4M) — EdTech shows the smallest absolute change of all sectors, +\$4.9M, and the flattest slope on the chart.
- **Quantified spread:** the gap between the best (CleanTech, +1093%) and worst (EdTech, +327%) performer is 766 percentage points, showing a 3.3x disparity in relative growth trajectory across sectors.

Q3. Time Series & Trends — STL Decomposition

Brief: A 48-month (Jan-2022 to Dec-2025) monthly total-funding series for the 6-sector pool was simulated (upward trend + annual seasonality + noise + one injected anomaly) and decomposed with STL (period=12).

```
trend = 60 + 1.3*t
seasonal = 16*sin(2*pi*(t-9)/12) # Q4 peak, mid-year trough
funding = trend + seasonal + normal(0,3,48)
funding[30] += 38 # one mega-round anomaly
stl = STL(ts, period=12, robust=False).fit()
```



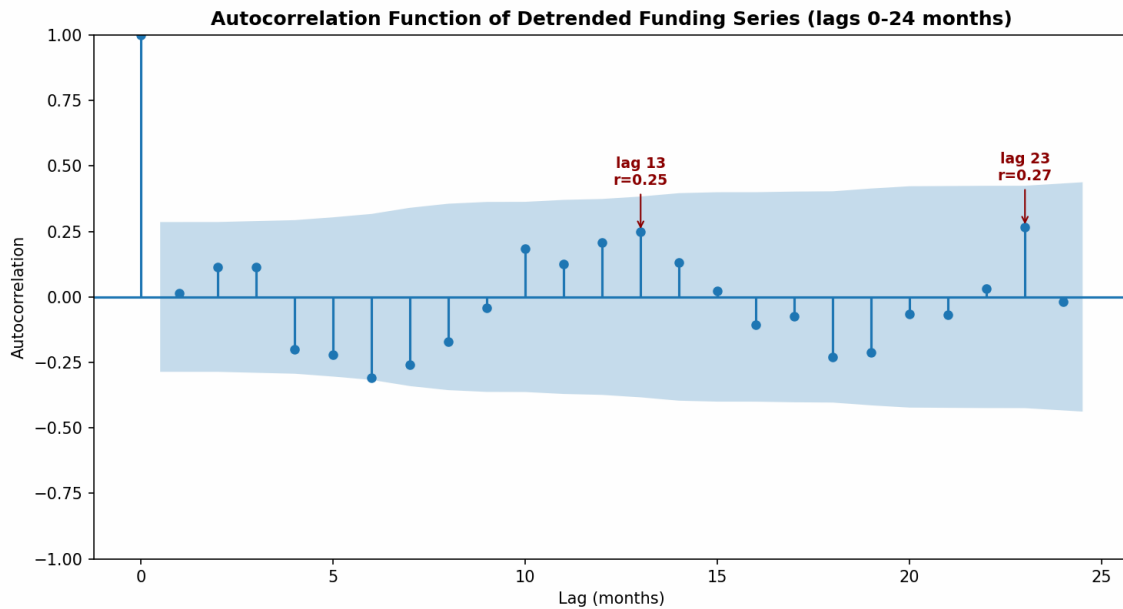
Analysis

- **Trend dominates:** trend strength $F_T = 0.945$ (near the 1.0 ceiling), rising steadily from \$60.1M (Jan-2022) to \$119.1M (Dec-2025) — roughly a 98% increase over 4 years at $\approx \$1.3\text{M}/\text{month}$.
- **Seasonality is also strong but secondary:** $F_S = 0.883 < F_T = 0.945$, confirming trend is the larger driver, though the seasonal swing itself is sizeable — amplitude (max–min) of \$44.3M, i.e. funding oscillates about $\pm 22\text{M}$ around trend within each year, peaking around Oct–Dec and dipping around Jun–Jul.
- **Flagged residual spike:** July 2024 shows a residual of +\$23.46M — 8.7x the typical residual standard deviation of \$2.69M (all other months) — clearly identifying that month as an anomaly (e.g., one mega-round) rather than part of the normal trend/seasonal pattern.

Q4. Time Series & Trends — Autocorrelation / Periodicity

Brief: Using the same monthly funding series (first-differenced to remove trend so the cycle is isolated), the ACF was computed out to 24 lags to detect the dominant cycle length.

```
ts = raw_ts.diff().dropna()      # detrend
acf_vals, confint = acf(ts, nlags=24, alpha=0.05)
plot_acf(ts, lags=24)
local_max = [i for i in 1..23 if acf[i]>acf[i-1] and acf[i]>acf[i+1]]
```



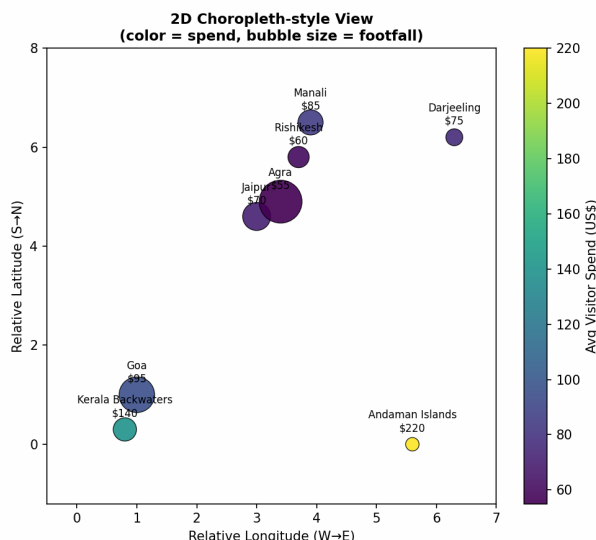
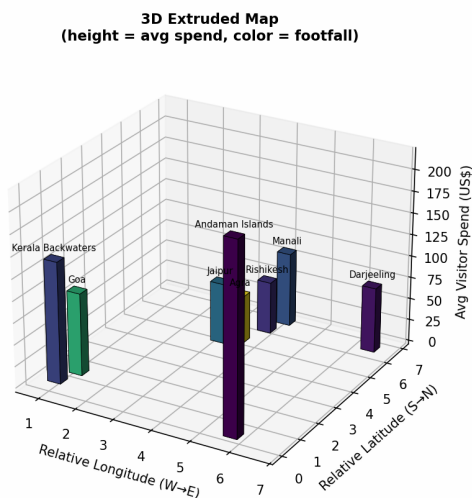
Analysis

- **Two peaks near the 1-year and 2-year marks:** the ACF shows local maxima at lag 13 ($r=0.25$) and lag 23 ($r=0.27$) — both sitting right next to the 12-month and 24-month marks, i.e. one and two years apart, which is the signature of an annual cycle.
- **Confirming trough structure:** between the two peaks the ACF dips to strongly negative values at lag 6 ($r=-0.31$) and lag 18 ($r=-0.23$) — troughs roughly half a cycle (6 months) away from each peak, exactly what is expected for a single ~ 12 -month periodic component rather than a shorter (e.g., quarterly) cycle.
- **Consistency with Q3:** the ~ 12 -month spacing between the lag-13 and lag-23 peaks ($\Delta \approx 10$ months, close to one year) corroborates the STL seasonal period=12 assumption and its $F_S=0.883$ finding — both diagnostics independently point to the same annual cycle.

Q5. Geospatial Visualization — 3D Extruded Map

Brief: A Tourism Board wants footfall vs visitor spend compared across 8 destinations. A 3D extruded bar map (bar height = avg. spend, colour = footfall, base = rough relative map position) was built alongside a 2D choropleth-style bubble view for comparison.

```
ax1 = fig.add_subplot(1,2,1, projection='3d')
ax1.bar3d(x, y, 0, dx, dy, avg_spend, color=viridis(footfall))
ax2 = fig.add_subplot(1,2,2)
ax2.scatter(x, y, s=footfall*1.8, c=avg_spend, cmap='viridis')
```



Analysis

- **Clear outlier reads well in 3D:** Andaman Islands stands isolated at the bottom-right ($x=5.6$, $y=0.0$) with by far the tallest bar (spend \$220, more than 2.5x the next highest) — its extreme height and isolation make it instantly identifiable in the 3D view.
- **But 3D hides close comparisons:** Jaipur (\$70 spend) and Agra (\$55 spend) sit only 0.4 units apart on the map ($x=3.0/3.4$, $y=4.6/4.9$); in the 3D view their bars are adjacent and near-identical in apparent height due to perspective foreshortening, making the true \$15 (27%) gap hard to read. The same crowding affects Rishikesh (\$60) and Manali (\$85), which sit only 0.2–0.9 units apart.
- **2D choropleth resolves it precisely:** in the 2D bubble view the same four destinations are unambiguous — Agra is visibly darker-purple ($\approx \$55$) than the lighter Jaipur ($\approx \$70$), and exact dollar labels remove any ambiguity, while bubble size still encodes footfall (Agra's bubble, footfall=610k, is visibly the largest on the plot).
- **Net effect:** the 3D extrusion improves comparison only for the single, spatially-isolated outlier (Andaman); for the 4 geographically clustered destinations (Jaipur/Agra/Rishikesh/Manali) it actively hinders accurate comparison versus the 2D choropleth/bubble view.